Standard Article Template

Testing the Native LaTeX Renderer

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1 Introduction

This document tests the native LaTeX renderer across a wide range of constructs. We include **colored text**, mathematics, tables with `booktabs`, code listings, custom theorem environments, and `tcolorbox` environments.

The renderer should handle all of the following:

I. Mathematical typesetting (inline and display)

II. Custom macro expansion

III. Color support via `xcolor`

IV. Table environments with `booktabs`

V. Code listings

VI. Box environments via `tcolorbox`

2 Mathematics

2.1 Inline Mathematics

Einstein's famous equation $E = mc^2$ appears inline. The quadratic formula gives $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$ for roots of $ax^2 + bx + c = 0$.

2.2 Display Mathematics

The Fourier transform of a function $f \in L^1(\mathbb{R})$ is defined as:

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x)\, e^{-2\pi i x \xi}\, dx \tag{1}$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \tag{2}$$

$$\nabla \cdot \mathbf{B} = 0 \tag{3}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{4}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{5}$$

Maxwell's equations (2)–(5) govern classical electromagnetism.

Definition 2.1 (Continuity)**.** *A function $f : \mathbb{R} \to \mathbb{R}$ is continuous at x_0 if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon$.*

Theorem 2.2 (Intermediate Value Theorem)**.** *If $f : [a, b] \to \mathbb{R}$ is continuous and y is between $f(a)$ and $f(b)$, then there exists $c \in (a, b)$ with $f(c) = y$.*

3 Tables

Table 1 shows a comparison using `booktabs` style rules.

Table 1: Performance comparison of rendering methods

Method	SSIM ↑	Time (ms) ↓	Memory (MB) ↓
CoreText basic	0.72	45	12
CoreText + Knuth-Plass	0.88	120	18
Full engine (ours)	**0.94**	**180**	**22**
Tectonic (reference)	1.00	3200	180

Table 2: A tabularx table with automatic column widths

ID	Description	Status	Value
1	This is a longer description that should wrap within the X column	Active	42
2	Another item with a description	Inactive	17
3	Third entry	Active	99

4 Color and Boxes

Text can be red, blue, or dark green. You can also use highlighted backgrounds.

Important Note

This is a `tcolorbox` environment. It supports colored frames and backgrounds, titles, and arbitrary content including mathematics: $\int_0^1 x^2\,dx = \frac{1}{3}$.

Warning

This renders a warning box. The `tcolorbox` package is widely used in modern LaTeX documents for highlighted content.

5 Code Listings

Listing 1: Main rendering function

```
func typeset(_ blocks: [TexBlock]) -> TypesetDocument {
    var pages: [TypesetPage] = [TypesetPage(pageNumber: 1)]
    var columnIndex: Int = 0
    var cursorY: CGFloat = geometry.marginTop

    for block in blocks {
        switch block {
        case .paragraph(let inlines):
            let attrStr = makeParaAttrStr(inlines, fonts: fonts)
            appendTextBlock(attrStr, pages: &pages, ...)
        case .mathDisplay(let src):
            let node = MathParser.parse(src)
```

```
            // ... render math
        default:
            break
        }
    }
    return TypesetDocument(pages: pages, ...)
}
```

Inline code: `TeXTypesetter.typeset(_:)` is the main entry point.

6 Lists with Enumitem

✓ First item with check mark

✓ Second item

✓ Third item

(c) This list starts at (c)

(d) Continues with (d)

(e) And so on

7 Cross-References

We defined the Fourier transform in Eq. (1). Maxwell's equations are in Eq. (2)–(5). The Intermediate Value Theorem is Theorem 2.2. Results are in Table 1.

8 Figures

Figure 1 illustrates the rendering pipeline. Images are loaded via `CGImageSource` and scaled to fit the column width.

9 Bibliography

Prior work by LeCun et al. [2] established deep learning as a general-purpose feature extractor. The attention mechanism [4] underpins modern large language models. Classical typesetting follows Knuth [1]. Online collaborative tools like Overleaf [3] have changed how papers are written.

10 Conclusion

This document exercises most of the packages and constructs needed for real-world academic papers. Any construct not rendered correctly represents a gap to close in the parity effort.

Figure 1: A sample figure demonstrating `\includegraphics` support. The image is automatically scaled to 70% of text width.

References

[1] Donald E. Knuth. *The TeXbook.* Addison-Wesley, 1984.

[2] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton. Deep learning. *Nature*, 521:436–444, 2015.

[3] Overleaf Team. Overleaf: Real-time collaborative writing and publishing. `https://www.overleaf.com`, 2023.

[4] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.